

CSC621 Data Structures and Algorithm

Lab Exercise 2 - Arrays

Question 1

You are required to write a program to use parallel arrays to determine grades of students from the given marks using the instructions given below.

Parallel arrays: is a data structure for representing arrays of records. Values located at the same index in each array are implicitly the fields of the same record.

Write a program to meet the following requirements:

1. Allow the user to get the students' IDs and their corresponding total marks, an integer, of 5 students in the course CSC621 from a standard input.
2. Store the data into two separate parallel arrays. One array should contain IDs and another array should contain marks.
3. Print out all the data on the screen after data entering is done.
4. Make a function `char determine_grade(int marks)` to determine students' grades. Use the following table for grade distribution.

Marks	Grade
80-100	A
65-79	B
50-64	C
0-49	D

5. Write a function `print_students_records` to print students' IDs and their corresponding marks and grades.
6. Determine the number of students who have passed CSC512 from your program by use of a function?

Question 2

Do the following changes to your previous exercise:

- i. Create a class for student so that it can encapsulate **marks**, **id** and **grade**. Make sure you have constructor, destructor and all setter and getter methods.
- ii. Instead of parallel arrays use just one array of student class for this program. Do all the required changes.

Modify function *void print_students_records* to pass an object of student instead of passing students' information separately. Syntax for the prototype of modified function should be *void print_students_record (student s[], int size)*.

Question 3

Delete ***char determine_grade(int marks)*** function and make it a behavior of student class. Your prototype should become ***void determine_grade()***. Better name for this method could also be ***void assign_grade()*** or ***void allocate_grade()***.

Question 4

Create a text file with student information, then read the information from the text file rather than the keyboard.